A Smaller Explicit Contraction Bound and Stronger Precision Barriers

A review of twenty SubexponentialSAT proposals,
rectangular matrix contraction, and asymmetric Hermite recovery

Research prepared for Vladimir Reshetnikov
with ChatGPT

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Abstract

We review ProveIt at commit `dde01c4a63b822c25b8eb345299346c36f837b49`: twenty SubexponentialSAT refinement packets, including five in the newest placement, and their two primary manuscripts. We obtain smaller explicit time upper bounds in both polynomial-space and cached regimes for the common normalized Boolean tensor-contraction problem. Let

$$\mathfrak{E}(s) = \frac{s}{\sqrt{\lceil \log_2 s \rceil}}, \qquad C_S = \frac{\pi}{\ln 2}\sqrt{\frac{2812500}{49997}}, \qquad C_T = \left(1 + \frac{1}{200000}\right) C_S.$$

Then, relative to the imported contraction identities and oracle interface, the new algorithm uses

$$(s+\mathcal{L})^{O(1)} 2^{(C_T + o(1))\mathfrak{E}(s)} \quad \text{time}, \qquad (s+\mathcal{L})^{O(1)} 2^{(C_S + o(1))\mathfrak{E}(s)} \quad \text{space},$$

where $C_T < 34$ and $C_S < 34$. A separate streamed composition gives $2^{(53.9775312 + o(1))\mathfrak{E}(s)}$ time with polynomial space, improving the latest stated 55.7094783 and a stronger compatible 55.4046659 composition. The strongest explicitly stated generic cached time coefficient in the reviewed corpus is $35.3545880584\ldots$. The improvement combines a genuine rectangular matrix bound with a fully charged multistage cache recurrence. It includes the cost of constructing every child and block table, and uses exact arithmetic over the actual power-of-two quotient rings.

This is an improvement of an exponential coefficient, not of the logarithmic order. We do not establish a smaller unrestricted SAT order than $n \log\log n/\sqrt{\log n}$, or close any of the conditional $n/\log n$ interfaces. To sharpen the search for such an improvement, we prove that arbitrary asymmetric Hermite multiplicities have the same leading independent-jet precision optimum, $(6 + o(1))M/u$, as the best symmetric taper. We also give an explicit inventory and dense-table barrier that incorporates the newer adaptive Smith separators. Exact rational certificates, finite algebraic checks, source identities, and integration notes accompany the article. No end-to-end SAT implementation or new formal verification is claimed.

Reading the result correctly. The requested unrestricted improvement in exponent *order* remains open in this work. The completed positive result is the strict improvement of the explicitly stated, directly comparable generic contraction-time certificates in both memory regimes. The negative results concern specified representations and precision certificates, not the complexity of SAT.

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1 Review cutoff and the comparison that is actually justified

The read-only review began at `f898a9fde4` and closes at ProveIt commit `dde01c4a63b822c25b8e b345299346c36f837b49`. The relevant chronology is:

1. `2b925d8800`: placement of the first seven refinement packets and the two primary manuscripts;
2. `3e09608991`: vendoring of the OpenAI preprint at upstream `7ef5c0cc5ca4b0e072b7a050fca0 9ab24bc6fa3a`;
3. `45233238ae`: a summary separating established refinements from conditional proposals;
4. `6d3adc4506`: placement of eight second-round packets;
5. `f42479bca2` and `f898a9fde4`: arrival of the four newer ZIPs reviewed here;
6. `450252e9d8`: placement of those four packets plus `sat_adaptive_contraction`, whose arrival is `fa6e904d10`. The closing head also merges unrelated work.

The initial committed summary did not include the four incoming packets. The closing check found the fifth new packet. All five are now placed, and the closing incoming directory contains only its index. We verified all 180 files in the four original extracted packets byte-for-byte against their new locations. We inspected the new LaTeX proofs as well as the relevant placed reports and the vendored quantitative source. The source index explicitly states that filing a packet did not verify its mathematics or run a formal build [1, 2, 3].

1.1 Three different meanings of strongest

For fixed clause width k, the strongest stated density envelope supported by the reviewed ordinary mathematical derivations is

$$\operatorname{poly}_k(n,m)\,2^{O_k\left(\frac{n}{\sqrt{\log n}}\,\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1.1}$$

It specializes to $2^{O_{k,D}(n/\sqrt{\log n})}$ for fixed density $m \le Dn$, and has unrestricted exponent $O_k(n\log\log n/\sqrt{\log n})$ [5, 6]. All twenty packets retain these worst-case orders wherever they state a proved global algorithm.

For *explicit coefficients on normalized tensor wires*, the strongest stated generic polynomial-space coefficient is $55.70947831891\ldots$ [16]. Combining that new streamed separator with the earlier multistage schedule gives the stronger derived comparator $55.40466590661\ldots$, which we also exceed. Allowing subexponential memory, the strongest stated generic time coefficient is $35.354588058401\ldots$, with space coefficient $34.274803005023\ldots$ [13, 15]. Another cached algorithm uses time $37.93010908288\ldots$ and smaller space $33.62304092166\ldots$. Thus there is a time-space portfolio, not one method that dominates every memory allowance.

For *conditional statements*, the strongest reviewed target is

$$\operatorname{poly}_k(n,m)2^{O_k(n\log\log n/\log n)}, \qquad \operatorname{poly}_{k,D}(n,m)2^{O_{k,D}(n/\log n)}$$

at unrestricted and fixed density, respectively. It requires an unconstructed structured-mode summation interface [7]. Other conditional proposals require compact factorization, degree-preserving recursion, or linear-wire compression [8, 9]. These implications are not available algorithms. Our coefficient improvement does not imply bounds smaller than these conditional targets.

1.2 The twenty packets

In the table, same order means that no smaller unrestricted SAT logarithmic order is established by that packet. Coefficients refer to the normalized wire scale defined in Section 2.

Packet	Relevant contribution	Limitation or dependency
`cauchy_mode_improvement`	Integral factorial Cauchy modes with asymptotic weight slope three.	Same order.
`eth_contraction_improvements`	Exact taper count, precision $6MD/u$, streamed extraction; coefficient 648.4411.	Imports the source contraction.
`3sat_precision_refinement`	Exact recovery loss and optimal symmetric real taper.	Faster structured summation is conditional.
`sat_density_improvement`	Uniform density envelope (1.1).	Worst-case and fixed-density orders unchanged.
`eth_density_refinement`	Related density envelope and precision savings.	Overlaps the preceding density result.
`3sat_refinement`	Scalar parity lift; a smaller incidence-degree cap.	Compact second-stage factorization missing.
`3sat_asymptotic_analysis`	Scoped barriers and precise degree/compression interfaces.	No completed stronger-order algorithm.
`eth_asymmetric_packet_balancing`	Asymmetric side-degree invariant; coefficient 76.9090.	Lower-degree 31.0641 has stronger premises.
`3sat_round2_refinement`	Matching-potential packing; coefficient 84.0663; actual-domain rank audit.	Same order.
`eth_balanced_packing`	Nearly uniform whole-vertex pair loads; 94.7482.	Same order.
`eth_high_moment`	Moment-based deterministic balancing; 94.7482.	Lower-degree specializations are separate.
`eth_matching_refinement`	Matching potentials and thin gaps; 94.7482.	Same order.
`eth_packet_refinement`	Near-unit padding and logarithmic guard; 95.6639.	Same order.
`eth_recursive_refinement`	Missing-residue modes of slope 64/21; 133.2074.	Same order.
`eth_joint_cauchy`	Full crossing-block slope four; cached contraction.	Its ingredients improve later compositions.
`sat_frontier_improvements`	Multistage polynomial-space coefficient 63.8574; cached 38.4845.	Lower-degree cached 19.0839 has stronger premises.
`subexponential_sat_frontier`	Polynomial-space 63.8522; cached 37.9301; precision peeling.	Same order.
`subexponential_sat_kernel_improvements`	Optimal integral Smith separation on actual ports; Gray, antipodal and projective extraction.	Instance-specific or lower-order kernel gains.
`sat_adaptive_contraction`	Streamed conditioning: 55.7095 polynomial space; adaptive cached 36.7503.	Lower-degree cached 15.3919 has stronger premises.
`subsat_contraction_tree`	Binary block-table construction; exact CRT bridge; cached time 35.3546.	Uses subexponential space.

The accompanying `audit/frontier_audit.md` gives exact source filenames and theorem labels

for every row. The first fifteen packets are available through the pinned project index [1]; the last five are also in the closing placement [12, 13, 14, 15, 16].

1.3 Limits of numerical domination

The separate manuscript credited to Claude has the same relevant n-only order but supplies no matching explicit coefficient in the normalized wire model. Consequently a numerical comparison with its unspecified constant is unavailable [4]. Similarly, an input-class-specific coefficient such as 15.3919 for degree at most $s^{1/12}$ must not be compared as if it applied to the generic degree-$s^{9/10}$ class. We retain those algorithms as options under their own premises.

A coefficient improvement also does not prove a lower bound on the older algorithm's runtime. What can be compared are the displayed certifying functions. For the new C_T and the previous explicit $C_{\mathrm{old}} = 35.354588058401\ldots$,

$$\frac{2^{(C_T+o(1))\mathfrak{E}(s)}}{2^{(C_{\mathrm{old}}+o(1))\mathfrak{E}(s)}} = 2^{-(C_{\mathrm{old}}-C_T+o(1))\mathfrak{E}(s)} \longrightarrow 0.$$

Polynomial prefactors do not change this conclusion. This is the precise sense in which the completed positive result is an asymptotically smaller upper-bound certificate.

2 The inherited contraction interface

We use a bipartite Boolean tensor network with at most s wires and a polynomial number of vertices. Each edge carries a bit. A vertex tensor is an implicitly specified function of its incident bits, and the contraction sums the product of these functions over all edge assignments. The initial entries lie in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \le P_0, \ell \le s.$$

The maximum degree is at most $\lceil s^{9/10} \rceil$. For fixed A, b, an original entry is computable in $(s+\mathcal{L})^b 2^{As^{9/10}}$ time and $(s+\mathcal{L})^b$ space, where $\mathcal{L}$ is the description length. Fixed constant multiples in the precision bounds are harmless; a change of the edge-budget normalization changes numerical coefficients and must be tracked.

Write $\lambda = \lceil \log_2 s \rceil$. At depth d, a call has M wires, group alphabet size r, packet length $L = 2^a$, and u retained valuation levels. The coefficient ring is enlarged to a polynomial-rank unramified extension and a truncated polynomial ring, exactly as in the source. Tensor descriptions preserve their creation-time representatives. We import the correctness of the source deformation, Hermite reconstruction, and factorization into child contractions [3].

2.1 Packing and precision

Let $Q = r^a$. The asymmetric packing result gives padded mass $(1 + o(1))M$ and child size $(2/r + o(1))M$ when, uniformly,

$$QR = o(M), \qquad Q^2 C\lambda^2 = o(M), \qquad LQ^2 = o(M). \tag{2.1}$$

Here R, C are the degree bounds on the two sides. At the root both are at most $\lceil s^{9/10} \rceil$. Below the root we orient the retained tensors on the first side and the newly created packet tensors on the other, so

$$R \le \lceil s^{9/10} \rceil, \qquad C \le 2^{a_{d-1}+1}.$$

This is the crucial asymmetric invariant [10].

Use

$$b = \lceil 6 \log_2 \lambda \rceil, \quad u_d = \lfloor f_d \lambda \rfloor, \quad a_d = \sum_{j=0}^{d} (u_j + b).$$

For a node with $i = v_2(t) < u_d$ put $D = 2^{a_d - i}$. The sharp taper and additive buffer imply the uniform working precision

$$p_t \le (6 + o(1)) MD/u_d. \tag{2.2}$$

The stored precision inherited from an ancestor is $O(s\lambda^2 2^{a_{d-1}})$. Since $D \ge 2^{a_{d-1}+b+1}$ and active calls have $M > s/\lambda$, its ratio to MD/u_d is $O(\lambda^4 2^{-b}) = o(1)$. The jet-evaluation correction is lower order as well. These are the same imported precision guarantees used by the newer frontier packets [7, 13, 15].

2.2 Full-block modes and a uniform entropy certificate

For every directed pair e of distinct groups, full crossing-block Cauchy separation supplies

$$(2 + o(1)) \frac{M}{r^2 D}$$

mode families with weights

$$a + 1 - i + 4q - 2s_2(q), \qquad q = 0, 1, \ldots,$$

with the appropriate finite endpoint. Source and target locality is preserved [11]. For each directed pair fix an exact total weight budget W_e, with $\sum_e W_e < p_t$, and let $\mathcal{I}_e$ be the list of mode tuples with that exact weight. Set $h_e = \log_2 |\mathcal{I}_e|$; empty lists contribute nothing and singleton lists have $h_e = 0$. For fixed r, there are only polynomially many budget arrays.

The common-tilt estimate in the incoming frontier and tree packets is

$$\sum_e z_e h_e \le \left(\frac{\sqrt{\sum_e z_e^2}}{r} + o(1) \right) K(M, u), \qquad K(M, u) = \kappa \frac{M}{\sqrt{u}}, \quad \kappa = \frac{\pi\sqrt{2}}{\ln 2}, \tag{2.3}$$

for any one of finitely many fixed nonnegative weight vectors z. It is important that this estimate also holds when individual W_e are small or zero.

For clarity, its constant comes from

$$\sum_{q \ge 0} -\ln(1 - e^{-\theta(1+4q-2s_2(q))}) = \frac{\pi^2 + o(1)}{24\theta}.$$

Use parameter θ/z_e for pair e, multiply its logarithmic generating-function bound by z_e, and sum. The leading expression is

$$\theta p_t + \frac{\pi^2}{24\theta} \frac{(2 + o(1))M}{r^2 D} \sum_e z_e^2.$$

Optimizing θ and substituting (2.2) gives (2.3). The thin buffer makes the remainders uniform. Thus we import a proven common upper bound, not an unjustified large-budget approximation for every individual list.

2.3 The cache recurrence must pay for child evaluation

For a fixed exact-weight array, cache each group function on its incident pair indices. An entry is a real child contraction. There are $2(r-1)$ incident directed pairs, so the cache-construction coefficient is

$$a_r = \frac{\sqrt{2(r-1)}}{r}.$$

If $b_r K(M,u)$ bounds local table construction and contraction, the correct recurrence, up to polynomial and uniform lower-order factors, is

$$T_d \le 2^{(a_{r_d}+o(1))K_d} T_{d+1} + 2^{(b_{r_d}+o(1))K_d}. \tag{2.4}$$

Treating the group functions as free tables would omit the first term and produce an invalid improvement. The next section sharpens b_r and supplies a schedule that controls the entire recurrence.

3 A smaller certified coefficient from rectangular multiplication

The latest incoming contraction-tree packet [15] uses a square matrix multiplication backend of exponent 12/5 and proves time coefficient $35.3545880584\ldots$ on $s/\sqrt{\log_2 s}$. Its polynomial-coordinate and Chinese remainder bridge already permits field algorithms over the actual coefficient ring. We replace its square-block rectangular bound by a genuinely rectangular algorithm, including its input/output cost, and change the retained-range schedule. No faster summation oracle is assumed. This improves the leading coefficient of the displayed exponential upper bound; the logarithmic order is unchanged.

3.1 The imported field facts and the exact ring interface

Only two established field inequalities are needed:

$$\alpha > \frac{3}{10}, \qquad \omega < \frac{19}{8}.$$

Here α is the dual matrix multiplication exponent. For example, *New Bounds for Matrix Multiplication: from Alpha to Omega*, arXiv:2307.07970 [17], gives stronger inequalities. No current-record assertion is needed. The paper formulates tensor rank over an arbitrary field in Sections 3.1–3.4; specialize the field to $\mathbb{Q}$. Fix $\varepsilon = 1/100000$. The definition and constructive proof of the dual exponent supply a fixed finite rational bilinear algorithm for multiplying dimensions $N \times N^{3/10} \times N$ with exponent at most $2+\varepsilon$. Choose a fixed square algorithm with exponent at most 19/8. Their dimensions and rational coefficients are constants independent of the SAT input. Padding to fixed powers changes dimensions only by constant factors; the bounds below allow those factors.

Lemma 3.1 (Rectangular cost with the output term retained)**.** *Let the logarithms to base two of the three matrix dimensions, in decreasing order, be $x \ge y \ge z \ge 0$. Matrix multiplication is possible in a polynomial factor times $2^{F(x,y,z)}$ bit operations over the coefficient rings of the normalized contraction interface, where*

$$F(x,y,z) = \max\left\{(1+\varepsilon/2)(x+y), \quad x + \left(\frac{47}{56}+\varepsilon\right)y + \frac{15}{28}z\right\}. \tag{3.1}$$

The workspace is a polynomial factor times the largest input or output array. The polynomial depends only on the fixed algorithms and on the normalized interface.

Proof. Put $\alpha_0 = 3/10$ and $\omega_0 = 19/8$. First suppose the two larger logarithms are equal to y. If $z \le \alpha_0 y$, pad the smaller dimension and use the rectangular algorithm, with exponent at most $(2+\varepsilon)y$. If $z \ge \alpha_0 y$, take the tensor product of the rectangular algorithm at logarithmic size

$$t_0 = \frac{y - z}{1 - \alpha_0}$$

and the square algorithm at logarithmic size

$$t_1 = \frac{z - \alpha_0 y}{1 - \alpha_0}.$$

Then $t_0 + t_1 = y$ and $\alpha_0 t_0 + t_1 = z$. Round the two recursion depths upward independently in their respective fixed base dimensions. Both padding ratios and the rank overhead are then bounded by constants uniformly in x, y, z, including a depth that is zero. The rank exponent is at most $(2+\varepsilon)t_0 + \omega_0 t_1$. Blocking the largest dimension contributes $x - y$, so both cases are bounded by

$$\max\left\{x + y, \quad x + \frac{47}{56}y + \frac{15}{28}z\right\} + \varepsilon y.$$

Since $y \le (x+y)/2$, this is at most (3.1). The first branch is essential: a field exponent $2+\varepsilon$ does not justify a free quadratic input/output branch. Permuting the three dimensions uses the symmetries of the matrix multiplication tensor; transposition or cyclic transposition of the fixed bilinear identities implements the corresponding product. Concretely, permute the three linear-form factors in the finite tensor-rank decomposition, regard the new third factor as output coefficients, and execute that bilinear decomposition afresh. We do not transpose a cached program and assume its memory schedule survives. This is an arithmetic algorithm, not a promise that only a scalar trace is required.

For completeness, the coefficient ring is a free module of polynomial rank over $\mathbb{Z}/2^p\mathbb{Z}$, with a polynomial-size integer multiplication table, and p is polynomial in s. In particular this includes

$$R = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell)$$

with the monic polynomials and degrees in the source interface. Lift each module coordinate to $[0, 2^p)$. For each ordered pair of coordinates, compute an ordinary integer matrix product. An unreduced scalar entry is less than $d_2 2^{2p}$, hence has polynomial bit length. Clear the finitely many constant denominators in the fixed field algorithms, omit their finitely many exceptional primes, and perform the product over enough remaining odd prime fields that their product exceeds this integer reconstruction bound. The number and bit lengths of these primes, and CRT reconstruction, are polynomial in s. Apply the specified integer structure constants and reduce modulo 2^p. Polynomially many coordinate pairs suffice. Thus no even integer or nilpotent is inverted in R.

A fixed bilinear algorithm can process one recursive product at a time and accumulate the output. At each level its input, output and auxiliary arrays are bounded by a constant times the largest input/output size, and recursive sizes decrease. The largest array size is the maximum of the three pairwise dimension products, which is invariant under those permutations. Tensor products and blocking preserve this bound. Coordinate pairs and CRT primes can also be processed sequentially. Storing residue information adds only polynomial bits per cell. Consequently the workspace is as stated, including the output array. Sequential record reblocking and sorting give the same leading bit-cost exponents without unit-cost random-access assumptions on exponentially large indices. □

3.2 The local entropy calculation

Import the exact pair-index factorization, asymmetric packing, thin precision gaps, and slope-four Cauchy modes from the reviewed packets. At a call with M wires and retained range u, write

$$K(M,u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}} = \kappa\frac{M}{\sqrt{u}}.$$

For an exact pair-budget array, let h_e be the logarithm of the number of mode tuples on the directed group pair e. The inherited common-tilt bound is, for every fixed nonnegative vector w,

$$\sum_e w_e h_e \le \left(\frac{\sqrt{\sum_e w_e^2}}{r} + o(1)\right) K(M,u). \tag{3.2}$$

There are only polynomially many exact budget arrays because all arities below are fixed. Counts and rank/unrank operations use the inherited polynomial dynamic program; an entry of a group function remains a real recursive contraction.

Put

$$d = 1 + \varepsilon/2, \qquad v = \frac{47}{56} + \varepsilon, \qquad w = \frac{15}{28}.$$

If a matrix operation has three disjoint classes of directed pairs with cardinalities $q_1 \ge q_2 \ge q_3$, then Lemma 3.1 and (3.2) give squared local coefficient, before dividing by r^2,

$$\Phi(q_1,q_2,q_3) = \max\{d^2(q_1+q_2),\ q_1 + v^2 q_2 + w^2 q_3\}. \tag{3.3}$$

Indeed sorting the actual logarithmic dimensions selects one of six permutations. For each permutation apply the weighted entropy estimate; the rearrangement inequality pairs the largest squared weights with the largest cardinalities for a uniform upper bound. This argument includes empty and small pair lists and makes no separate asymptotic assumption on individual pair budgets.

Build each block table by balanced binary merges and finish by contracting three blocks as a triangle. A merge of group sets of sizes a, b, with $k = a + b$, has class cardinalities

$$2a(r-k), \quad 2ab, \quad 2b(r-k).$$

For final block sizes k_1, k_2, k_3, the three classes have cardinalities $2k_1k_2, 2k_1k_3, 2k_2k_3$. The following complete maxima include every merge, the final multiplication, and all input/output tables:

r	final block sizes	cache cardinality	largest boundary cardinality	$\max\Phi$
8	$(3,3,2)$	14	30	$30d^2$
10	$(4,3,3)$	18	48	$48d^2$
6	$(2,2,2)$	10	16	$16d^2$
4	$(2,1,1)$	6	8	$8d^2$

These are rational inequalities, checked exactly by the accompanying script. For instance the root triangle has classes $(18,12,12)$ and

$$18 + 12v^2 + 12w^2 < 30 < 30d^2.$$

A size-three root merge has classes $(20,10,4)$ and also costs at most $30d^2$. The input/output branch therefore dominates the root certificate. Dropping it would falsely suggest a coefficient below the table-size cost.

Define

$$a_r = \frac{\sqrt{2(r-1)}}{r}, \qquad (j_8, j_{10}, j_6, j_4) = \left(\frac{\sqrt{30}}{8}, \frac{\sqrt{48}}{10}, \frac{2}{3}, \frac{1}{\sqrt{2}}\right), \qquad b_r = dj_r.$$

The time recurrence is

$$T_i \le 2^{(a_{r_i}+o(1))K_i} T_{i+1} + 2^{(b_{r_i}+o(1))K_i} \tag{3.4}$$

up to controlled polynomial factors. Thus every cached child entry is paid for. The local table-space exponent is $(j_{r_i} + o(1))K_i$.

3.3 A strict rational schedule

Let $\lambda = \lceil \log_2 s \rceil$, stop recursion at $H = \lceil s/\lambda \rceil$, and use the arities and retained-range fractions

$$\begin{aligned} &(r_0, r_1, r_2) = (8, 10, 6), \qquad r_{3+j} = 4 \quad (j \ge 0), \\ &f_0 = \frac{49997}{3000000}, \qquad f_1 = \frac{67}{5000}, \qquad f_2 = \frac{17}{2000}, \\ &f_{3+j} = \frac{34003}{8000000}\left(\frac{5}{8}\right)^j. \end{aligned}$$

Put $u_i = \lfloor f_i \lambda \rfloor$, $b = \lceil 6 \log_2 \lambda \rceil$, and $a_i = \sum_{t=0}^{i} (u_t + b)$. The leading cumulative fractions are

$$A_0 = \frac{49997}{3000000}, \quad A_1 = \frac{90197}{3000000}, \quad A_2 = \frac{115697}{3000000}, \quad A_\infty = \frac{499}{10000}.$$

The root condition is $2A_0 \log_2 r_0 + 9/10 < 1$, while subsequent old-side conditions are $A_i \log_2 r_i + 9/10 < 1$. The exact integer comparisons

$$10^{500} < 2^{1661}, \qquad 6^{200} < 2^{517}$$

give the following rational upper exponents, all strictly below one:

regime	old-side exponent bound
0	499997/500000
1	1499817217/1500000000
2	599815349/600000000
$i \ge 3$	4999/5000

At later calls the fresh-side degree is at most $2^{a_{i-1}+1}$, so the remaining packing exponents are bounded by $2A_i \log_2 r_i + A_{i-1}$ and $(2 \log_2 r_i + 1)A_i$; both are below one with ample fixed margin. New packet degrees are $s^{0.0499+o(1)}$, below the maintained $s^{9/10}$ bound. Choose padding tolerance $O(\lambda^{-2})$; the strict power gaps absorb this tolerance, all polylogarithmic factors, and the additive buffers.

For a noncircular depth check, provisionally verify these estimates through $3 + \lceil \log_{20/11} \lambda \rceil + O(1)$ active levels. Throughout this range the smallest taper satisfies

$$u_i \ge \lambda^{\eta+o(1)}, \qquad \eta = 1 - \frac{\ln(8/5)}{\ln(20/11)} > 0,$$

because $5/8 > 11/20$. Floors are uniformly negligible. Nearly lossless packing gives children with at most $11M/20$ wires eventually, since every arity is at least four; hence no active path exceeds the provisional depth. This closes the bootstrap. Accumulated padding errors tend to zero, giving

$$M_0 \le s, \quad M_1 \le (1+o(1))s/4, \quad M_2 \le (1+o(1))s/20, \quad M_{3+j} \le (1+o(1))\frac{s}{60\,2^j}.$$

The buffer gives $D \ge 2^{a_{i-1}+b+1}$. The inherited precision $O(s\lambda^2 2^{a_{i-1}})$ is $O(\lambda^4 2^{-b}) = o(1)$ times MD/u_i, so the sharp coefficient-six Hermite precision and entropy calculation remain valid at all active calls.

3.4 The resulting time and space bounds

For $i = 0, 1, 2, 3$, put

$$(m_0, m_1, m_2, m_3) = (1, 1/4, 1/20, 1/60), \qquad K_i^* = \frac{\kappa m_i}{\sqrt{f_i}}.$$

Subsequent coefficients decline with ratio $\gamma = (1/2)/\sqrt{5/8} = \sqrt{2/5}$. Moreover $a_4/(1-\gamma) > b_4$. Unrolling the complete cache recurrence, including all child calls, gives the maximum of

$$\begin{aligned} &b_8 K_0^*, \\ &a_8 K_0^* + b_{10} K_1^*, \\ &a_8 K_0^* + a_{10} K_1^* + b_6 K_2^*, \\ &a_8 K_0^* + a_{10} K_1^* + a_6 K_2^* + \frac{a_4 K_3^*}{1-\gamma}. \end{aligned}$$

Exact rational interval evaluation encloses these respectively by

$$33.993891, \qquad 32.812822, \qquad 31.412630, \qquad 33.657386.$$

The root is strictly largest. Ancestor and descendant tables occupy a sum of geometrically decreasing array sizes, so their space exponents take a maximum. The corresponding coefficients are

$$33.993721, \qquad 9.590642, \qquad 2.317442, \qquad 1.158670,$$

and again the root is largest.

Theorem 3.2 (A coefficient below 34)**.** *Under the normalized tensor interface and imported identities stated above, there is a deterministic exact contraction algorithm with*

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_T+o(1))s/\sqrt{\log_2 s}}, \qquad S(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_S+o(1))s/\sqrt{\log_2 s}},$$

where

$$\begin{aligned} C_S &= \frac{\pi\sqrt{2}}{\ln 2}\frac{\sqrt{30}}{8\sqrt{49997/3000000}} = 33.9937208906\ldots, \\ C_T &= \left(1+\frac{1}{200000}\right) C_S = 33.9938908592\ldots < 34. \end{aligned}$$

Consequently 34 is a valid eventual time coefficient and a valid eventual space coefficient. The space use is subexponential, not polynomial.

Polynomial interpolation and budget factors across $O(\log \lambda)$ levels contribute $O(\lambda \log \lambda)$ exponent bits. The numbers of recursive branches and all multiplicity polynomials depend on s alone; description-length processing is charged once per visited call. Thus the factor in $\mathcal{L}$ remains a fixed polynomial, rather than being raised to the recursion depth. Direct leaves cost $2^{O(s/\lambda)}$, the permitted original entry routines cost $2^{O(s^{9/10})}$, and new packet entry routines cost $\exp(s^{0.0499+o(1)})$. All are lower order after the displayed recurrence charges. Immutable creation-time representatives retain the source oracle accounting. No input, intermediate or output numerical precision grows beyond a polynomial in s.

The new time coefficient is about 3.85% below the newest delivered $35.3545880584\ldots$. A stronger comparator is the old square-block method retuned to its root-eight capacity boundary, namely $35.0635988462\ldots$; the improvement over that comparison is about 3.05%. Neither comparison asserts global optimality over every old schedule. For a fixed common normalization, a strict coefficient decrease $\delta > 0$ divides the displayed upper bound by $2^{(\delta+o(1))s/\sqrt{\log_2 s}}$. It does not identify the actual runtime ratio. These are coefficients per normalized wire, and cannot be substituted for hidden coefficients per SAT variable.

4 A polynomial-space coefficient below 54

A closing repository check added the packet `sat_adaptive_contraction` [16]. Its recursive-conditioning result changes the polynomial-space comparison frontier. The factor $\sqrt{17/24}$ in that packet reduces the number of paid child evaluations for streamed summation at arity sixteen; it is not a uniform improvement of the pair entropy bound and cannot multiply the cached coefficient from Section 3. We instead compose its proved streamed identity with a new multistage retained-range schedule.

4.1 The appropriate composed comparator

The new packet's displayed polynomial-space coefficient is $55.70947831890997\ldots$. Combining its conditioning identity with the previously filed `subexponential_sat_frontier` schedule [13] already gives the stronger bound $55.40466590661055\ldots$. That prior schedule has arities $(13, 14, 8, 6, 4, 4, \ldots)$ and fractions

$$\frac{27}{2000}, \quad \frac{51}{4000}, \quad \frac{141}{20000}, \quad \frac{107}{20000}, \quad \frac{27}{6400}\left(\frac{5}{8}\right)^{j} \quad (j \geq 0).$$

We use the stronger composed comparator below. Neither it nor our new schedule is asserted to optimize all possible prior parameters.

4.2 Stream exact pair lists using balanced separators

Retain the exact pair-budget arrays and the inherited entropy scale

$$K(M, u) = \kappa \frac{M}{\sqrt{u}}, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

Fix a balanced binary tree on the r groups. At a split $B = B_1 \sqcup B_2$, enumerate the pair indices crossing the split, recursively sum the two sides conditional on these indices, and multiply the conditional answers. Ancestor indices remain fixed parameters. Finite distributivity proves the

identity: after fixing crossing indices, no factor in one side depends on an internal index of the other side. A singleton evaluates its group factor by a genuine recursive tensor contraction.

The sets of cut pairs on a root-to-leaf path are disjoint. If their union is E_ℓ, the number of paid group-factor evaluations is at most

$$\sum_{\ell=1}^{r} \prod_{e \in E_\ell} |\mathcal{I}_e|.$$

Sibling internal sums add to this work rather than multiplying through a leaf evaluation. The largest cardinality of E_ℓ is

$$v(1) = 0, \qquad v(r) = 2\lfloor r/2 \rfloor \lceil r/2 \rceil + \max\{v(\lfloor r/2 \rfloor), v(\lceil r/2 \rceil)\}.$$

Thus, with

$$\beta(r) = \frac{\sqrt{v(r)}}{r},$$

the inherited subset-entropy estimate gives the fully paid recurrence

$$T_i \le \mathrm{poly}(s)\, 2^{(\beta(r_i)+o(1))K_i} T_{i+1}. \tag{4.1}$$

This is the new packet's proved separator identity, applied at the arities below. It uses no field multiplication algorithm or cached exponential table. The separator assignment ranks, finite-ring accumulators and recursion frames have polynomial total size. All expensive group evaluations remain explicitly charged.

4.3 A strict multistage schedule

Let $\lambda = \lceil \log_2 s \rceil$, $H = \lceil s/\lambda \rceil$, $b = \lceil 6 \log_2 \lambda \rceil$, and

$$u_i = \lfloor f_i \lambda \rfloor, \qquad a_i = \sum_{t=0}^{i} (u_t + b).$$

Choose

$$(r_0, r_1, r_2, r_3, r_4) = (16, 16, 8, 6, 5), \qquad r_{5+j} = 4 \quad (j \ge 0),$$

with

$$f_0 = f_1 = \frac{12499}{1000000}, \qquad f_2 = \frac{2083}{250000}, \qquad f_3 = \frac{107}{20000},$$
$$f_4 = \frac{219}{50000}, \qquad f_{5+j} = \frac{513}{200000}\left(\frac{5}{8}\right)^j.$$

The leading cumulative capacities at the first five levels and in the tail are exactly

$$0.012499, \quad 0.024998, \quad 0.033330, \quad 0.038680, \quad 0.043060, \quad A_\infty = 0.049900.$$

All displayed decimals are terminating rational numbers, not numerical approximations used in place of inequalities.

The root high-degree packing constraint is $9/10 + 2A_0 \log_2 r_0 < 1$. At later calls the old-side constraint is $9/10 + A_i \log_2 r_i < 1$. Using the exact integer comparisons

$$6^{200} < 2^{517}, \qquad 5^{500} < 2^{1161},$$

the six regime bounds are, respectively,

$$\frac{124999}{125000}, \quad \frac{124999}{125000}, \quad \frac{99999}{100000}, \quad \frac{4999939}{5000000}, \quad \frac{24999633}{25000000}, \quad \frac{4999}{5000}.$$

Every one is strictly below one. The fresh-side and packet-padding exponents are bounded by $(2\log_2 r_i + 1)A_i < 1/4$ in all six regimes. New degrees are at most $s^{0.0499+o(1)}$, below the maintained $s^{9/10}$ bound. Consequently all packing gates tolerate the additive buffers and any fixed polylogarithmic slack.

As before, first check all hypothetical active levels through $5 + \lceil \log_{20/11} \lambda \rceil + O(1)$. The smallest taper in this range is at least $\lambda^{\eta+o(1)}$, with

$$\eta = 1 - \frac{\ln(8/5)}{\ln(20/11)} > 0.$$

All arities are at least four, so nearly lossless padding implies eventual child shrinkage at most $11/20$, closing the depth bootstrap. Padding may be chosen with relative error $O(\lambda^{-2})$, whose accumulation tends to zero over this depth. The sharper limiting wire masses are

$$(m_0, m_1, m_2, m_3, m_4, m_5) = \left(1, \frac{1}{8}, \frac{1}{64}, \frac{1}{256}, \frac{1}{768}, \frac{1}{1920}\right), \qquad m_{5+j} = m_5\, 2^{-j}.$$

The six cut profiles are

$$(v(16), v(16), v(8), v(6), v(5), v(4)) = (170, 170, 42, 24, 18, 10).$$

The same precision guard gives inherited-to-local precision ratio $O(\lambda^4 2^{-b}) = o(1)$, preserving the coefficient-six precision budget and the common-tilt entropy estimate uniformly.

4.4 Exact summation of the new certificate

Unroll (4.1). With $\gamma = (1/2)/\sqrt{5/8} = \sqrt{2/5}$, the resulting coefficient is

$$C_{\mathrm{ps}} = \kappa \left[\sum_{i=0}^{4} \frac{m_i \beta(r_i)}{\sqrt{f_i}} + \frac{m_5 \beta(4)}{\sqrt{f_5}(1-\gamma)} \right] = 53.97753119806873\ldots. \tag{4.2}$$

Its six contributions are approximately

$$46.7203678473, \quad 5.8400459809, \quad 0.8888315523, \quad 0.2794968539, \quad 0.1070058158, \quad 0.1417831479.$$

The accompanying standard-library script verifies the profile recursion, wire masses, capacities, every strict packing gate, and rational interval bounds for all square roots, π, and $\ln 2$. It proves

$$53.97753119806 < C_{\mathrm{ps}} < 53.97753119808 < 54 < 55.40466590660.$$

These finite interval checks certify the arithmetic in the all-size proof; they do not stand in for the imported contraction identities.

Theorem 4.1 (Polynomial-space multistage conditioning)**.** *Under the normalized tensor interface and imported identities, the above streamed algorithm has*

$$T(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(C_{\mathrm{ps}}+o(1))s/\sqrt{\log_2 s}}, \qquad C_{\mathrm{ps}} = 53.97753119806873\ldots,$$

and polynomial space. In particular, 54 is a valid eventual time coefficient.

The proof's remaining costs are unchanged: direct leaves take $2^{O(s/\lambda)}$, initial entry routines take $2^{O(s^{9/10})}$, and fresh packet routines take $\exp(s^{0.0499+o(1)})$. All are lower order after charging the ancestor separator enumerations. Multiplicity polynomials depend on s alone and contribute $O(\lambda \log \lambda)$ to the exponent across the active depths. A single fixed polynomial in the description length pays for processing at each visited call. Depth-first separator summation and tensor recursion retain polynomial space throughout.

The new coefficient is about 3.11% below the newest packet's displayed $55.7094783189\ldots$, and about 2.58% below the stronger prior composition $55.4046659066\ldots$. It complements the cached bound below 34 by improving the polynomial-space alternative. Both retain the same logarithmic order; neither proves the unresolved $n/\log n$-type targets or supplies a numerical coefficient per SAT variable.

5 Asymmetric Hermite nodes have the same precision optimum

The earlier exact minimax theorem treats valuation-symmetric multiplicities [7]. An earlier barrier explicitly leaves asymmetric node selection open [9]. We remove that restriction for the *fixed Hermite reconstruction with independent jet errors*. This is a precision-certificate theorem. It does not assert that the jets attainable by the source deformation can vary independently, or rule out recovery that exploits their correlations.

5.1 Exact loss of a node's recovery block

Let $a \geq u \geq 1$, $N = a + 1$, $L = 2^a$, and let S be a complete integer residue system modulo 2^N containing zero. For example, $S = \{-L, \ldots, L-1\}$. Assign nonnegative integer multiplicities q_t to the nonzero nodes, put $q_0 = 0$, and require $q_t = 0$ when $v_2(t) \geq u$. There is no symmetry or monotonicity requirement.

For an active node t, put $i = v_2(t)$, $D_t = 2^{a-i}$ and

$$E_t(T) = \prod_{v \in S \setminus \{0,t\}} \left(\frac{T-v}{t-v}\right)^{q_v}.$$

The ordinary Hermite reconstruction of $F(0)$ uses the first q_t coefficients of $F(t + 2^N z)$. The coefficient of its highest raw jet is exactly

$$E_t(0)(-t/2^N)^{q_t-1}.$$

Indeed, $E_t(t + 2^N z)^{-1}$ is an integral power series with constant term one. The top raw jet only contributes through that constant term. The negative valuation of the displayed coefficient is

$$\Lambda_t(q) = \sum_{v \neq 0,t} q_v(v_2(t-v) - v_2(v)) + (q_t - 1)(N - i). \tag{5.1}$$

Every other raw-jet coefficient is a sum of terms $E_t(0)c_j(-t/2^N)^h$ with c_j integral and $h \leq q_t - 1$. Its loss is therefore at most $\Lambda_t(q)$. The highest jet attains this bound exactly. Consequently

$$p + \max\{0, \Lambda_t(q)\}$$

is the exact nondecreasing precision allowance for this node's independent-error recovery block at output precision p. Define its normalized worst loss by

$$H(q) = \max\left(0, \max_{t:q_t>0} \frac{\Lambda_t(q)}{D_t}\right). \tag{5.2}$$

For $1 \le k \le N$, define the residue-subtree mass

$$S_k(t) = \sum_{\substack{v \in S \\ v \equiv t \pmod{2^k}}} q_v.$$

Expressing valuations as sums of congruence indicators gives

$$\Lambda_t(q) = \sum_{k=i+1}^{N} \big(S_k(t) - S_k(0)\big) - (N-i). \tag{5.3}$$

To verify the endpoint carefully, for $k < N$ excluding $v = t$ subtracts $(N-1-i)q_t$ from the valuation-difference sum. The highest-jet term adds $(q_t - 1)(N-i)$, leaving $q_t - (N-i)$. The q_t is precisely the $k = N$ subtree term.

5.2 Inactive nodes and convexity

Equation (5.1) can be evaluated as an affine expression even at inactive nodes. We also allow nonnegative real q_t as a relaxation; real multiplicities are not actual jet counts.

Lemma 5.1. *For every such nonnegative vector,*

$$H(q) = \max\left(0, \max_{\substack{t \in S\setminus\{0\} \\ v_2(t)<u}} \frac{\Lambda_t(q)}{D_t}\right).$$

In particular the relaxed objective is convex.

Proof. Within an annulus $v_2(t) = i$, both D_t and the negative terms in (5.3) are constant. Its maximization therefore depends only on the positive path sum $\sum_{k=i+1}^{N} S_k(t)$. If an inactive leaf has an ancestor containing positive mass, take the smallest such ancestor and an active leaf inside it. The two paths agree above that ancestor. Below it the inactive path has zero mass and the active path eventually has positive mass. The active leaf has a strictly larger path sum. Thus a nonempty annulus attains its maximum at an active leaf. An empty annulus has zero positive path mass and strictly negative affine loss. Adjoining zero makes its contribution irrelevant. The displayed expression is a maximum of affine functions. □

Theorem 5.2 (Symmetrization of arbitrary node multiplicities)**.** *Replace every q_t by the average multiplicity on its valuation annulus, obtaining $\overline{q}$. Then*

$$\sum_t \overline{q}_t = \sum_t q_t, \qquad H(\overline{q}) \le H(q).$$

The real minimum of this precision certificate is therefore attained by a valuation-symmetric profile, even when arbitrary asymmetric node choices are allowed.

Proof. Consider automorphisms of the complete binary residue tree that fix the leaf zero. They preserve $v_2(t)$ and $v_2(t-v)$, and act transitively on every annulus. Permuting the multiplicity vector by any such automorphism preserves (5.2) and total mass. Averaging over this finite group gives $\overline{q}$. Convexity from Lemma 5.1 proves the inequality. Only the valuations of the Hermite coefficients are used. We do not assume that the source polynomial deformation itself is invariant under these tree automorphisms. □

5.3 An exact finite minimax formula

For a symmetric profile, write q_i for its common value at valuation i and set $q_i = 0$ for $i \geq u$. Let $D_i = 2^{a-i}$ and $n_i = N - i$. The exact normalized loss is

$$\frac{\Lambda_i}{D_i} = \left(2 - \frac{1}{D_i}\right) q_i - \sum_{j=1}^{u-1-i} \frac{j}{2^j} q_{i+j} - \frac{n_i}{D_i}. \tag{5.4}$$

For instance, the finite normalization sum has diagonal $2 - (n_i + 1)/D_i$; adding $(q_i - 1)n_i/D_i$ gives precisely (5.4).

Given required normalized total mass $W > 0$, define backward rational recurrences

$$r_i = \frac{1 + \sum_{j=1}^{u-1-i} j2^{-j} r_{i+j}}{2 - D_i^{-1}}, \tag{5.5}$$

$$c_i = \frac{n_i/D_i + \sum_{j=1}^{u-1-i} j2^{-j} c_{i+j}}{2 - D_i^{-1}}, \tag{5.6}$$

$$R_* = \sum_{i=0}^{u-1} 2^{-i} r_i, \qquad C_* = \sum_{i=0}^{u-1} 2^{-i} c_i. \tag{5.7}$$

The starred C_* here is a local minimax quantity, unrelated to the algorithmic constants C_T, C_S.

Theorem 5.3 (Finite optimum and integer enclosure)**.** *Over all nonnegative real node multiplicities supported on $v_2(t) < u$ and satisfying $\sum_t q_t \geq LW$,*

$$\min H(q) = \eta_* = \max\left\{0, \frac{W - C_*}{R_*}\right\}.$$

If $W \geq C_*$, a minimizing symmetric profile is $q_i = \eta_* r_i + c_i$. If $W < C_*$, use $q_i = (W/C_*)c_i$. The optimum over integer multiplicities, even allowing arbitrary node-by-node asymmetry, lies in $[\eta_*, \eta_* + 2)$.

Proof. Theorem 5.2 reduces the real lower bound to symmetric profiles. If $H(q) \leq \eta$, equation (5.4) gives

$$q_i \leq \frac{\eta + n_i/D_i + \sum_{j=1}^{u-1-i} j2^{-j} q_{i+j}}{2 - D_i^{-1}}.$$

Backward induction yields $q_i \leq \eta r_i + c_i$. Since an annulus contains $D_i = L2^{-i}$ nodes, completeness requires $W \leq \sum_i 2^{-i} q_i \leq \eta R_* + C_*$. This proves the stated lower bound. The displayed profile achieves equality in every constraint if $W \geq C_*$. Scaling c_i downward when $W < C_*$ makes every affine loss nonpositive and gives exact mass W.

For the integer upper bound, round every component of the chosen symmetric real profile upward. The total mass cannot decrease. In (5.4), the own-coordinate increase is less than $2 - D_i^{-1} < 2$, while every other increase has a nonpositive coefficient. Thus H rises by less than two. The unrestricted integer minimum is no smaller than its real relaxation, which proves both sides. □

5.4 The leading constant six survives asymmetry

Let $D_{\min} = 2^{a-u+1}$. In the limit $D_i = \infty$, the backward recurrence has the explicit solution, already implicit in the symmetric taper analysis [7],

$$r_i^{(\infty)} = \frac{u-i}{6} + \frac{5}{18} + \frac{2}{9}\,4^{-(u-i)}, \qquad R_\infty = \frac{u}{3} + \frac{2}{9} - \frac{2}{9}\,4^{-u}.$$

Positivity of the finite recurrences gives

$$R_\infty \le R_* \le \left(1 - \frac{1}{2D_{\min}}\right)^{-u} R_\infty, \qquad 0 \le C_* \le \frac{a+1}{D_{\min}} R_*. \tag{5.8}$$

For the first estimate, expand the backward recurrence as a sum over increasing index paths. Each visited index changes the limiting contribution by a factor at most $(1 - 1/(2D_{\min}))^{-1}$, and a path visits at most u indices. The second estimate follows by comparing the inhomogeneous terms in (5.5) and (5.6).

Hermite recovery for degree at most $M(2L-1)$ requires total multiplicity at least $M(2L-1)+1$, so

$$W = \frac{M(2L-1)+1}{L} = (2 + o(1))M.$$

If $u \to \infty$, $u/D_{\min} \to 0$, $(a+1)/D_{\min} = o(M/u)$, and $M/u \to \infty$, equations (5.8) and Theorem 5.3 imply

$$\min_{\text{integer node profiles}} H(q) = (6 + o(1))\frac{M}{u}. \tag{5.9}$$

The thin-gap schedules have $D_{\min} \ge \lambda^6$ and $a, u = O(\lambda)$, so these conditions hold at active calls. Arbitrary omission or asymmetric weighting of nodes within the retained valuation levels therefore cannot improve the leading MD/u precision scale in this fixed independent-error certificate.

The exact finite checker tests the valuation identity, the inactive-node extension, symmetrization, and the minimax construction. Its role is to detect finite algebraic mistakes; the general theorem is proved above rather than inferred from those checks.

6 An inventory barrier that survives adaptive integral separation

This result combines the exact full Cauchy Smith profile already proved [11] in `eth_joint_cauchy` with the pointwise optimality theorem [14] in `subexponential_sat_kernel_improvements`. The added conclusion also covers the sizes of explicit exact-weight tables. It is a lower bound on specified representations, not a lower bound for SAT or arbitrary contraction algorithms.

Let $\mathcal{O}$ be the ring of integers of an unramified finite extension of $\mathbb{Q}_2$, and put $R_P = \mathcal{O}/2^P\mathcal{O}$. For an odd unit τ, the integral Cauchy block

$$C_{jk} = \frac{2^\delta}{\tau + 2(j-k)}, \qquad 0 \le j, k < J,$$

has Smith valuations

$$e_j = \delta + 4j - 2s_2(j),$$

truncated when they reach P. Here $s_2(j)$ is binary popcount. One proof applies the Cauchy determinant formula: an h-minor has valuation

$$h\delta + h(h-1) + v_2\prod_{j<k}(x_k - x_j) + v_2\prod_{j<k}(y_k - y_j).$$

Every Vandermonde on h distinct integer points is divisible by $\prod_{j=0}^{h-1} j!$, and consecutive indices attain equality. The successive determinantal-ideal valuations yield the displayed profile. The full actual crossing orientation at valuation level i has the same profile with multiplicity 2^i and $\delta = a + 1 - i$; this follows from its integral triangular factorization with unit diagonal. This latter fact is imported from the cited packet, not claimed as new here.

Lemma 6.1 (Integral separated profiles are dominated by Smith profiles)**.** *Suppose a matrix over R_P has its nonzero Smith valuations sorted as $e_1 \le \cdots \le e_t$, and admits an integral separated expression*

$$K = \sum_{j=1}^{N} 2^{w_j} a_j b_j^{\mathsf{T}}, \qquad 0 \le w_1 \le \cdots \le w_N < P.$$

Then $N \ge t$ and $w_h \le e_h$ for $1 \le h \le t$.

Proof. For each $1 \le q \le P$, reduce modulo 2^q. The image of K has exactly $\#\{h : e_h < q\}$ nonzero cyclic summands, hence that many minimal generators. It is contained in the module generated by the $\#\{j : w_j < q\}$ surviving target columns. Over a finite chain ring, the minimum number of generators equals the residue-field dimension of the socle, and inclusion induces inclusion of socles. Consequently

$$\#\{h : e_h < q\} \le \#\{j : w_j < q\}.$$

The case $q = P$ gives $N \ge t$. If $w_h > e_h$, the choice $q = e_h + 1$ contradicts the last inequality. This is the optimality argument of the incoming adaptive-separation theorem. □

Theorem 6.2 (Explicit binary inventory lower bound)**.** *Let A, D, J, δ, P be positive integers, let $u \ge 1$ be real, and put $M = AD$. Suppose*

$$D \ge 16\sqrt{u}, \qquad \delta \le \frac{D}{\sqrt{u}}, \qquad J \ge \left\lfloor \frac{D}{8\sqrt{u}} \right\rfloor, \qquad P \ge \frac{AD^2}{4u}.$$

For the direct sum of A copies of the preceding Cauchy block, every integral separated expression with guaranteed weights w_j has at least

$$2^{M/(16\sqrt{u})}$$

formal binary mode tuples satisfying both $\sum_j w_j \beta_j < P$ and $\sum_j \beta_j \le M$.

Proof. Set $q = \lfloor D/(8\sqrt{u}) \rfloor$. Since $D/(8\sqrt{u}) \ge 2$,

$$q \ge \frac{D}{16\sqrt{u}}.$$

The first Aq Smith weights consist of the first q weights in each copy. Their sum is bounded by

$$\begin{aligned} A \sum_{j=0}^{q-1} (\delta + 4j - 2s_2(j)) &\le A\big(q\delta + 2q(q-1)\big) \\ &\le \frac{5AD^2}{32u} < P. \end{aligned}$$

In particular these Smith entries are all nonzero at precision P. Lemma 6.1 shows that the first Aq weights of any integral separated expression have no larger sum. Every one of their 2^{Aq} binary subsets is therefore feasible. Its ordinary degree is at most $Aq \le AD = M$. The lower bound for q completes the proof. □

Corollary 6.3 (An exact-weight list and a dense table must be large)**.** *Under the hypotheses of Theorem 6.2, at least one integer b with $0 \le b < P$ has an exact-weight tuple list of size at least*

$$\frac{2^{M/(16\sqrt{u})}}{P}.$$

An algorithm that explicitly materializes a group-function table with one cell for every member of this list must write at least that many cells.

Proof. Partition the binary tuples from the theorem according to their integer total weight. There are at most P possible weights. Pigeonholing gives the first claim, and explicit materialization gives the second. □

For a fixed number of groups, some directed group pair carries a fixed positive fraction of any family of filled crossing blocks. Applying the corollary to that pair gives $2^{\Omega(M/\sqrt{u})}/P$ formal indices. In the reviewed schedules, $P = s^{O(1)}$, $u = O(\log s)$, the active wire count is at least $s/\operatorname{polylog} s$, and the buffer ensures $D/\sqrt{u} \gg \delta$. Thus adaptive integral basis selection, valuation-only pruning, the global ordinary-degree cap, and faster matrix multiplication on fully materialized exact-weight tables cannot by themselves remove this representation's square-root logarithmic scale. At the root $M = \Theta(s)$, it is $2^{\Omega(s/\sqrt{\log s})}$.

When each full packet orientation is separated individually, the argument also respects its row-degree cap: selecting q modes in each of its $R = 2^i$ copies uses $Rq \le L/2$ modes, where $L = RD$. If bases mix different packets, only the asserted global degree cap follows without additional support information.

No nonzero-entry lower bound is being asserted. Distinct formal tuples may cancel, yield zero, or have a compact joint representation. The bound does not constrain an algorithm that proves such cancellations, compresses the group tables, bypasses their explicit construction, or uses a different deformation. It also does not prove that every SAT encoding realizes the displayed filled family. These are precisely the ways a structural improvement could evade this local barrier.

7 An arity-uniform barrier for the full-table certificate model

The following result identifies a limit of the certificate used above, not a lower bound for SAT. It is useful because changing grouping arities or choosing a contraction tree adaptively might otherwise appear to evade the weighted-partition obstruction.

Consider the abstract leading entropy relaxation on the $r(r-1)$ directed pairs of $r \ge 3$ groups,

$$\mathcal{H}_r(K) = \{h \in \mathbb{R}_{\ge 0}^{r(r-1)} : \|h\|_2 \le K/r\}.$$

This is the norm-ball envelope corresponding to the common-tilt Cauchy–Schwarz certificates; it intentionally forgets more detailed realizability constraints of actual mode lists. A full table for a subset B of k groups has logarithmic size

$$L_B(h) = \sum_{e \in \partial B} h_e, \qquad |\partial B| = 2k(r-k).$$

A full-table binary contraction tree stores the prescribed table for each merged subset. The model includes the binary table-building schemes and their final triangle contraction used in the reviewed packets.

Proposition 7.1 (Fixed and adaptive table certificates)**.** *For every fixed full-table binary tree $\mathcal{T}$,*

$$\sup_{h\in\mathcal{H}_r(K)} \max_{B\in\mathcal{T}} L_B(h) \geq \frac{2}{3}K.$$

Even if the tree is chosen after h is known,

$$\sup_{h\in\mathcal{H}_r(K)} \min_{\mathcal{T}} \max_{B\in\mathcal{T}} L_B(h) \geq \frac{4}{9\sqrt{1-1/r}}K \geq \frac{4}{9}K.$$

Proof. Every full binary tree on r leaves has a node with leaf count $k \in [r/3, 2r/3]$. One proof starts at the root and descends while a child has more than $2r/3$ leaves. At the last such node its children both have at most $2r/3$ leaves, and at least one has at least $r/3$ leaves. For that node, $2k(r-k) \geq 4r^2/9$. The support function of the nonnegative norm ball gives

$$\sup_{h\in\mathcal{H}_r(K)} L_B(h) = \frac{K}{r}\sqrt{2k(r-k)} \geq \frac{2}{3}K,$$

proving the fixed-tree statement.

That argument cannot interchange the supremum over h with the minimum over adaptively chosen trees. For the adaptive statement instead use the single vector

$$h_e = \frac{K}{r\sqrt{r(r-1)}} \quad \text{for every directed pair } e.$$

It belongs to $\mathcal{H}_r(K)$ and works simultaneously against every tree. At each tree's balanced node,

$$L_B(h) \geq \frac{4r^2}{9}\frac{K}{r\sqrt{r(r-1)}} = \frac{4K}{9\sqrt{1-1/r}}.$$

□

The reviewed root packing certificate requires

$$2a_0 \log_2 r + \frac{9}{10}\log_2 s \leq (1-o(1))\log_2 s, \qquad u_0 \leq a_0.$$

Consequently, within that certificate,

$$u_0 \leq \left(\frac{1}{20}+o(1)\right)\frac{\log_2 s}{\log_2 r}, \qquad K(s,u_0) = \Omega\left(\frac{s\sqrt{\log r}}{\sqrt{\log s}}\right).$$

Proposition 7.1 therefore prevents a worst-case $o(s/\sqrt{\log s})$ certificate from full-table tree selection, arity retuning, or a faster matrix backend alone. The constants are uniform in r. This remains an obstruction in the idealized real entropy model even if one grants growing arities and ignores their extra budget-enumeration costs.

The statement is deliberately scoped. It does not claim that every point of $\mathcal{H}_r(K)$ is realized by an actual crossing network, that every instance materializes a table of this size, or that the packing inequality is necessary for all conceivable packing algorithms. It is a lower bound on this relaxed certificate model. Additional constraints on real mode lists, a nonmaterializing summation method, or a different packing or SAT representation may evade it. It supplies no complexity lower bound for SAT or for general tensor contraction.

The newly filed streamed separator branch. The recursive-conditioning certificate in Section 4 also retains this order, by a separate elementary calculation. Its root cut alone gives

$$v(r) \geq 2\lfloor r/2 \rfloor \lceil r/2 \rceil \geq \frac{r^2-1}{2}, \qquad \beta(r) = \frac{\sqrt{v(r)}}{r} \geq \sqrt{\frac{1-r^{-2}}{2}} \geq \frac{2}{3} \quad (r \geq 3).$$

Consequently its local subset-entropy certificate is a positive uniform multiple of $K(s, u_0)$ even when the arity changes. This does not extend the full-table proposition to all streaming algorithms; it verifies the same order obstruction for this particular balanced recursive-conditioning profile. An exact summation method exploiting additional structure remains outside both scoped arguments.

8 Why the exponent order is still unresolved

The positive contraction theorem decreases explicit coefficients in $2^{(C+o(1))\mathfrak{E}(s)}$. It does not replace $\mathfrak{E}(s)$ by a smaller order. The new Hermite theorem and the representation barriers identify several changes that cannot, on their own, supply that replacement.

8.1 Precision peeling does not justify a free reset

The incoming frontier packet already proves precision peeling: a mode tuple of total guaranteed weight $W < P$ can contribute as 2^W times a product of normalized child contractions modulo 2^{P-W}. It also supplies the required creation-time precision caps [13]. This identity is prior work, not a new contribution here.

A tempting further step is to reset the cumulative packet exponent because most tuples have small residual precision. The present upper-bound analysis does not justify that step. If its partition certificate at budget P is $S(P) = c\sqrt{P}$ with the other parameters fixed, then for any fixed $B > 0$,

$$S\left(P - \frac{P}{(\log s)^B}\right) = S(P)\sqrt{1-(\log s)^{-B}} = (1-o(1))S(P).$$

Thus the high-residual part, $P - W \geq P/(\log s)^B$, still has the same leading counting certificate. Meanwhile

$$\log_2(P-W) \geq \log_2 P - O(\log\log s).$$

It retains the same leading inherited precision exponent. That change can recover an additive $O(\log\log s)$ in the packet exponent, not a new range of order $\log s$. Typical-residual reasoning or a split into a few fixed fractional bins misses this obstruction. The calculation concerns the current certificate; it is not a lower bound on actual evaluation times. A successful reset would require collective handling or new structure in that high-residual part.

8.2 Why binary tuple indices do not close the recursive interface

The exact-weight mode lists can be ranked and unranked with polynomial work. This gives a compact bit string describing a mode tuple, but a group function at fixed grouping arity can still depend on a linear fraction of all the new bits. Its degree then exceeds the $s^{9/10}$ entry-oracle premise of the contraction theorem at that new size. Evaluating such an entry recursively does not split its ports at low wire cost.

The conditional structured-summation and degree-preserving proposals correctly isolate this missing factorization [7, 8, 9]. Applying the contraction theorem a second time without constructing it would be circular. The finite rank/unrank dynamic program is necessary bookkeeping, not the missing algorithm.

8.3 Research questions with a concrete success criterion

1. **Construct a small representation of a group function.** For a compressed mode network with B index bits, exhibit an exact network with $O(B)$ wires, sublinear maximum degree, and entry evaluation cost compatible with the second contraction. The representation must encode the entire group function, including coefficient extraction, not just its tuple indices. This is the most direct route to the existing $s/\log s$ conditional target.

2. **Exploit correlations among attainable Hermite jets.** Theorem 5.3 optimizes independent perturbations of the ordinary recovery blocks. It leaves open the lattice of jet vectors actually produced by the deformed contraction. Identify a uniform relation in that lattice and an exact recovery algorithm whose precision analysis uses it. Moving multiplicities among nodes alone now has a precise negative answer.

3. **Avoid materializing formal exact-weight tables.** The inventory barrier does not prove that their entries are independent, nonzero, or information-theoretically incompressible. A representation that supports multiplication and summation without constructing every formal cell could evade it. The Cauchy kernel's structure must be shown to survive contraction with arbitrary allowed vertex tensors; that survival cannot be assumed.

4. **Remove incidence entropy in the dense reduction.** The reviewed reduction uses $O(n)$ selected incidences among n polylog n possible incidences, leading to $O(n \log\log n)$ index wires. A direct $O(n)$-wire representation with the required oracle bounds would improve the unrestricted SAT order even if the contraction's square-root logarithm stayed unchanged. Polynomial degree in incidence variables does not automatically imply low complexity in the binary rank bits.

5. **Optimize time and space together under explicit premises.** The new rectangular and streamed algorithms can coexist with the lower-degree and lower-memory branches. Further schedule or hybrid separator/cache optimization may yield smaller coefficients. Such a result should be reported as a coefficient improvement unless it actually changes the logarithmic scale.

No new affirmative algorithm for these open interfaces is supplied in this packet. The established refinements, their imported dependencies, and the still-open stronger target are recorded separately in `PROOF_STATUS.json`.

9 Validation, dependencies, and reproducibility

The proof has three distinct layers. The source deformation, SAT reductions, packing theorem, full Cauchy separation, and existing separator identity are imported and cited. The new rectangular aggregation, explicit schedules, asymmetric symmetrization and finite minimax formula, and scoped barrier extensions are proved in this report. Exact computations check the finite arithmetic and identities on which those proofs depend.

9.1 Exact checks that were executed

Artifact	Executed checks
`code/check_schedule.py`	Rational rectangular-operation ledgers; all strict packing gates; four complete time candidates; space maxima; interval certificate $C_T < 34$.
`code/check_streamed.py`	Balanced separator profiles, wire masses, all capacities and packing gates, the convergent tail, comparison compositions, and the rational interval certificate $C_{\mathrm{ps}} < 54$.
`code/check_hermite.py`	2,318 exhaustive multiplicity vectors on grids of sizes 2, 4 and 8; subtree-loss identity, inactive-node extension, symmetrization; 112 finite rational minimax instances.
`code/check_inventory.py`	4,699 exact minor-valuation cases, 94 independent rational determinants, 768 inventory parameter cases, and 324 explicit binary tuples with exact-weight pigeonhole checks.

The schedule certificates use rational enclosures from Machin's formula for π, a positive series for $\ln 2$, and integer square roots. Floating-point values are presentation fields only; every acceptance comparison uses exact arithmetic. Runtime guards remain active under optimized Python. Independent review checked the rectangular interpolation, the nonfield CRT bridge, transposition, workspace accounting, complete recursion, and the Hermite symmetrization. The independent audit is included.

The four initially incoming packets were compared to their final placed locations: all $37 + 48 + 58 + 37 = 180$ files matched byte-for-byte. The provenance directory records the initial and closing commits, relevant commit messages, Git blob identities, and SHA-256 identities. The file inventory is an identity ledger, not a claim that every included source file received a line-by-line mathematical audit.

9.2 What these checks do not establish

This archive is not a new formalization of the upstream ETH argument. No Lean or Rocq build is claimed. No complete subexponential SAT solver, fast rectangular field-multiplication implementation, or benchmark of full SAT solving is included. The field exponent facts are imported mathematical results [17]; the code checks the exact interface arithmetic and resulting schedules, not those large finite algorithms.

Likewise, a rational certificate for an eventual asymptotic constant does not give a practical input-size threshold. The strict packing gaps are small and the inherited framework is asymptotic. The report therefore claims neither practical speedups nor that the displayed upper bounds are attained by prior algorithms.

9.3 Replay and integration

Python 3.10 or newer suffices for all included mathematical checks. From the archive's top-level directory:

```
python3 verify_manifest.py
python3 run_checks.py
python3 build_report.py
```

The last command requires pdfLaTeX and the packages named in the source. It disables shell escape and builds three passes. No command downloads sources or executes repository scripts. Regenerating results changes those output files, so verify the delivered manifest before replaying.

The complete article is `report.tex`, with modular section sources. The PDF is compiled from that source. The proposed integration location and changes to the repository index are described in `INTEGRATION.md`. This task produces an archive for review; it does not modify the remote repository. The proof-status file distinguishes new proofs, imported results, checks, and unresolved stronger targets.

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